

"Math Is Cool" Masters — 2025-26

5th grade — May 16, 2026

Sponsored by: WEA - Eastern

GENERAL INSTRUCTIONS applying to all tests:

- *Good sportsmanship is expected throughout the competition by all involved (competitors and observers). Display of poor sportsmanship will result in disqualification.*
- *Competitors may not use calculators or any other aids on any portion of this contest.*
- *Unless stated otherwise:*
 - *All answers are integers, and any non-integer answers will be "coded" as integers.*
 - *For 5th grade and up, all fractions and ratios must be reduced to simplest form, all radicals must be simplified, and all denominators must be rationalized.*
 - *Do not round or approximate answers. Leave answers in terms of π or other irrational quantities (e.g., $\sqrt{2}$), where applicable.*
- *Units are not necessary as part of your answer, However, if you choose to use units, they must be correct.*
- *Record all answers on the colored cover sheets in the answer column only.*
- ***Be sure that the student name, school, team number, etc. has been filled out at the top of each answer sheet.***
- *Tests will be scored as a 0 if answers are not recorded correctly on the answer sheets.*
- *Blank answer sheets and answer sheets with no name will be scored as a 0.*

FINAL SCORES AND AWARDS

Individual awards are determined by both the Mental Math and Individual Test scores. Individual ties are broken based on the following, in this order: total scaled individual points, total number of correct answers on the Individual Test, Mental Math raw score, number of correct answers from Individual Test #31-40, number of correct answers from Individual Test #16-30, highest numbered question answered correctly on the Individual Test working backwards from #40.

Team (School) awards are based on the highest score from amongst each of the school's "teams of 4 students" in each event and is calculated as $2 \cdot (\text{Sum of highest 3 Mental Math scores}) + 2 \cdot (\text{Multiple Choice}) + 6 \cdot (\text{Team}) + 1 \cdot (\text{Triple Jump}) + 1 \cdot (\text{College Bowl})$, for approximate weights of 25%, 20%, 30%, 15% and 10% respectively. Team ties are broken based on highest event score in order of the events, starting with Mental Math.

MENTAL MATH TEST - 30 sec./quest., 8 problems, ~8%/25% of individ./team scores

The proctor will read each question twice. You may not do any writing or talking while arriving at a solution. Record only your answer on your answer sheet. You may not change, cross out, erase, or write over an answer once you have written it down. The maximum wait time is 30 seconds after completion of the second reading of the question. Correct answers receive 1 point.

INDIVIDUAL TEST - 35 minutes, 40 problems, ~92% of individual score

When you are prompted to begin, tear off the colored answer sheet and begin testing. No talking during this individual test. You will be given a 5 minute time warning. Correct answers receive 2 points for problems 1-30 and 3 points for 31-40 (in the scaled score).

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Final Score (out of 8)

Room #

School Name

Student Name

Team #

Mental Math - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

STUDENT: DO NOT WRITE IN SHADED REGIONS (or anywhere else, other than the answer box)

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
5 th grade	TOTAL:		

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Key

Mental Math Contest - Answer Key

30 seconds per question - ~25% of team score & ~8% of individual score

SCORERS — Write-overs, Cross-outs, and Erasures Must be Marked Incorrect (0)
Bracketed items [...] in the answer key are optional.

5th grade

Answer		
1	12 [meters]	In meters, what is the perimeter of a square with an area of nine square meters?
2	300 [calories]	Playing ping pong for one hour burns one hundred fifty calories. How many calories would you burn playing ping pong for one hundred twenty minutes?
3	16 [= range]	What is the range of the following set: twelve, seventeen, fifteen, twenty-eight
4	5	What is the third smallest positive integer factor of one hundred five?
5	6 [ways]	A hot air balloon took off with a sheep, a duck, and a rooster on board. However, they had to board one at a time. In how many different orders could the animals have boarded the balloon?
6	3 [=x]	Solve this equation for x: Three X plus two X plus five plus seven equals twenty-seven
7	72 [dogs and cats]	The ratio of dogs to cats at an animal shelter was seven to five. After twelve dogs and zero cats were adopted out, the ratio of dogs to cats is now one to one. How many total dogs and cats were at the shelter originally?
8	8800 [yards]	A lion's roar can be heard from five miles away. From how many yards away can a lion's roar be heard?

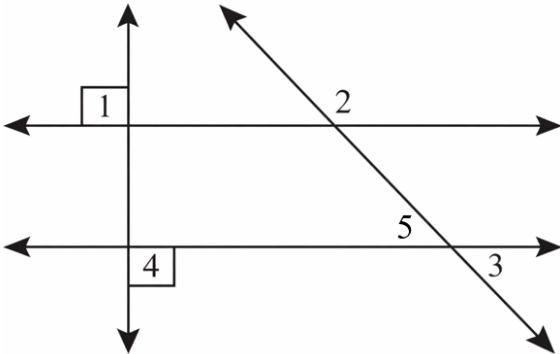
"Math Is Cool" Masters — 2025-26

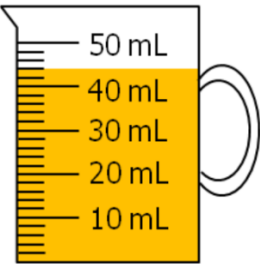
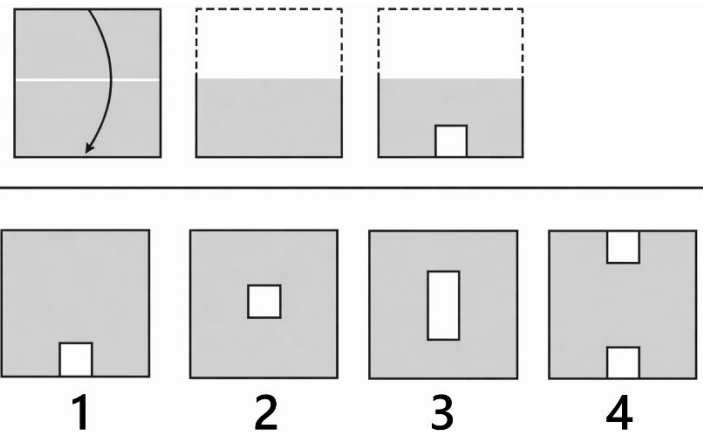
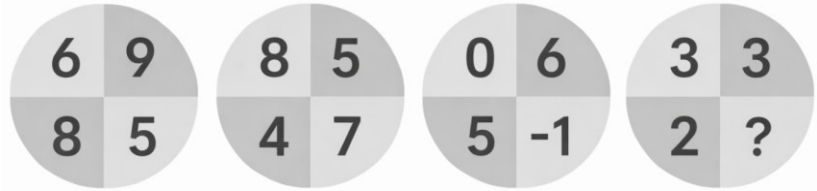
5th grade — May 16, 2026

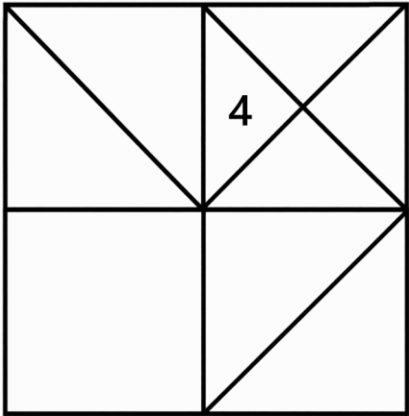
Individual Contest

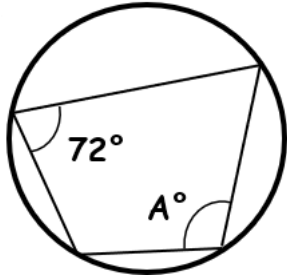
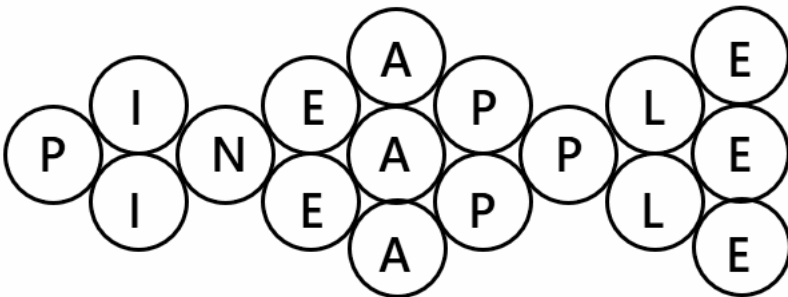
Record all answers on the colored cover sheet. 35 minutes, 40 problems, ~92% of individual score.

No talking during this individual test. A 5-minute time warning will be given.

	Questions 1-30: 2 points each
1	What is the smallest positive even integer?
2	What is the next number in the sequence 2, 6, 18, _
3	Nine feet is equal to how many yards?
4	The Earth is approximately 4.5 billion years old. When this number is written out as an integer, how many zeros are at the end of the number?
5	There are 6 people ahead of Xinyin in a line, and 6 people behind the person directly behind Xinyin in the same line. How many total people are in the line?
6	Ten percent of people are left-handed. If one person is randomly selected, what is the probability in percent that they are not left-handed?
7	Pound cakes got their name because the original recipe contained a pound each of butter, sugar, flour, and eggs. What percent, by weight, of a pound cake is flour?
8	There are about 1800 steps to the top of the Eiffel tower. If Jorge walks up them at a constant rate of 18 steps per minute, how many minutes would it take for him to get to the top?
9	Queen Elizabeth II had a personal net worth of \$425 million, which included the \$65 million Sandringham House and the \$140 million Balmoral Castle. How many million dollars of her net worth was not part of these two castles?
10	What is 11^2 ?
11	Which angle is obtuse? Your answer should be the angle number: 1, 2, 3, 4 or 5 
Continued on next page.	

12	Santa Claus was issued a pilot's license by the U.S. government in 1927! However, he does have to conduct a pilot review every two years for the rest of eternity, starting in 1929. In what year will he have his 100 th pilot review?
13	How many milliliters (mL) of liquid are contained in this measuring cup? 
14	What is the area of a parallelogram, in square meters, with base length 12 meters and height 5 meters?
15	Evaluate: $7345 - 765$
16	A Statistics class has twenty 15-year-olds, nineteen 14-year-olds, and one 34-year-old (the teacher). What is the mode, in years, of the ages of the people in this class?
17	A piece of paper is folded in half, and then a hole is punched through it. After the paper is unfolded again, which image number will it look like? Your answer should be the image number: 1, 2, 3 or 4.  1 2 3 4
18	About 7000 languages are spoken in the world. Approximately 12% of the world's total languages are spoken in Papua New Guinea. How many languages are spoken in Papua New Guinea?
19	What number, when multiplied by 6, is equal to 30?
20	In this number pattern, what number should replace the question mark? 
21	Evaluate: $2\left(\frac{18}{4+5} + 3\right) + 6$
22	Cooking oil boils at 230 degrees Celsius (C). Degrees Fahrenheit (F) can be found from degrees C as follows. What is the boiling temperature of cooking oil in degrees F? $F = \frac{9}{5}C + 32$
Continued on next page.	

23	A large square is divided into 4 smaller squares, and then some diagonals are drawn. One small triangular shape has an area of 4 square units. What is the total area of the large square in square units? 
24	The world record for stuffing drinking straws into your mouth at once is 650. Emilia really wants to break this record, so she starts training. On the first day, she manages to cram 100 drinking straws in her mouth, and every day after that, she increases the number of straws by 12. On what number day will she break the record?
25	A triangle has side lengths 6, 8, and 10. What is the degree measure of its largest angle?
26	In a round of musical chairs, there are 4 people and 3 chairs remaining, colored red, green, and blue. How many ways are there for people to sit down at the end of the round?
27	A Statistics class has twenty 15-year-olds, nineteen 14-year-olds, and one 34-year-old (the teacher). What is the mean number of years of the ages of the people in this class?
28	China has made a number panda-shaped solar farms in order to get more young people like you interested in renewable energy! When a certain number of solar panels are arranged in rows of 7 per row, there are 2 left over. When arranged in rows of 9 solar panels per row, there is 1 left over. When arranged in rows of 10 solar panels per row, there are none left over. What is the fewest number of solar panels in this group?
29	Chloe got 3 likes on her TikTok video, Adalynn got 1 like on her TikTok video, and Yesenia got 25 likes on her TikTok video. How many likes did Luca get on his TikTok video?
30	May 29 is "Put a Pillow on Your Fridge Day." To prepare, Wrenly buys one pillow on May 1, two pillows on May 2, continuing this pattern until she buys 28 pillows on May 28. On May 29, instead on buying any more pillows, she puts them all on her fridge. How many pillows does Wrenly put on her fridge?
Challenge Questions: 3 points each	
31	What is the smallest 5-digit positive integer in which three of the digits are distinct (different) odd integers and the other two digits are distinct even integers?
Continued on next page.	

32	How many times does the number 842 appear in the figure, when read from left to right with the digits in consecutive order? Find the number 842 <table><tr><td>2</td><td>7</td><td>8</td><td>4</td><td>4</td><td>2</td><td>2</td><td>3</td><td>8</td><td>7</td></tr><tr><td>0</td><td>2</td><td>0</td><td>3</td><td>9</td><td>5</td><td>4</td><td>4</td><td>6</td><td>2</td></tr><tr><td>7</td><td>6</td><td>0</td><td>6</td><td>3</td><td>8</td><td>4</td><td>9</td><td>8</td><td>3</td></tr><tr><td>5</td><td>3</td><td>6</td><td>3</td><td>3</td><td>7</td><td>4</td><td>1</td><td>6</td><td>9</td></tr><tr><td>6</td><td>0</td><td>9</td><td>8</td><td>8</td><td>0</td><td>8</td><td>4</td><td>2</td><td>3</td></tr><tr><td>3</td><td>6</td><td>5</td><td>9</td><td>6</td><td>1</td><td>9</td><td>7</td><td>9</td><td>9</td></tr><tr><td>8</td><td>4</td><td>1</td><td>5</td><td>9</td><td>2</td><td>5</td><td>9</td><td>2</td><td>1</td></tr></table>	2	7	8	4	4	2	2	3	8	7	0	2	0	3	9	5	4	4	6	2	7	6	0	6	3	8	4	9	8	3	5	3	6	3	3	7	4	1	6	9	6	0	9	8	8	0	8	4	2	3	3	6	5	9	6	1	9	7	9	9	8	4	1	5	9	2	5	9	2	1
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8	4	1	5	9	2	5	9	2	1																																																														
33	A quadrilateral is inscribed in a circle. What is the value of angle A in degrees? 																																																																						
34	Starting from the third term, a sequence follows the formula $a_n = 2a_{n-1} + a_{n-2}$, where a_n is the nth term. In other words, every term, starting from term 3, is equal to twice the term before it added to the term before that. If the sequence starts 0, 1,..., what is the 7th term?																																																																						
35	On the coordinate plane, a circle with radius 2 is centered at the point (0, 1). In how many points will it be intersected by the line $y = x - 2$?																																																																						
36	In how many different ways can the word PINEAPPLE be read from this diagram, starting at the left-hand P, and only moving to the right, to circles that are tangent to (touching) the current circle. 																																																																						
37	How many times do the minute and hour hand overlap on an analog clock in one day, starting at midnight (12:00 AM) and ending at 11:59 PM that same day?																																																																						
38	The first 6 terms of an infinite sequence are 1, 2, 2, 3, 3, 3, and so on, where each number is repeated the number of times equal to its value. What is the 100th term in this sequence?																																																																						
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39

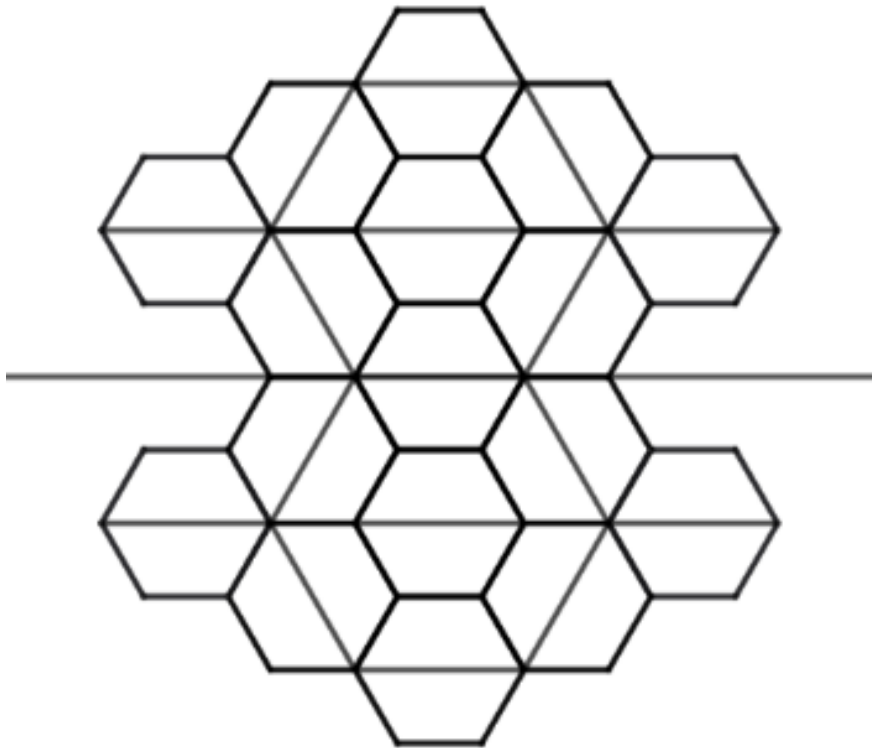
Julius built a neural network to recognize handwritten digits (the numbers 0 to 9). For each input handwritten digit, it outputs a digit as a guess. However, there are a few problems.

Any number with a "closed loop" (4, 6, 8, 9, and 0) gives an output that is that number subtracted from 9 (for example, 4 results in an output of 5). Any number that has an open loop (2, 3, 5, and 7) results in an output that is that number subtracted from 7 (for example, 3 results in an output of 4). The number 1 always results in an output of 1.

If Julius inputs two handwritten digits into the neural network in order to get a two-digit output that does not start with 0, what is the smallest number that this output could not be?

40

Each half of this figure is composed of 4 red trapezoids, 6 blue trapezoids and 7 white trapezoids. When the upper half is folded down over the centerline, 2 pairs of red trapezoids coincide, as do 4 pairs of blue trapezoids. There are 2 blue-white pairs. How many white pairs coincide?



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KEY

Individual Contest – Answer Key

SCORERS: Bracketed [...] items in answer key are optional. Just mark the score as 0 or 1 and add up those values to reflect total correct.

First Scorer – use the right-hand columns so 2nd scorer can do a blind scoring.

	Answer
1	2
2	54
3	3 [yards]
4	8 [zeroes]
5	14 [people]
6	90 [%]
7	25 [%]
8	100 [minutes]
9	220 [million dollars]
10	121
11	2 [angle #]
12	2127 [year]
13	44 [mL]
14	60 [square meters]
15	6580

	Answer
16	15 [years]
17	4
18	840 [languages]
19	5
20	2
21	16
22	446 [deg F]
23	64 [sq. units]
24	47 [day no.]
25	90 [degrees]
26	24 [ways]
27	15 [years]
28	100 [panda-shaped solar farms]
29	12 [likes]
30	406 [pillows]

	Answer
31	10235
32	2 [times]
33	108 [degrees]
34	70
35	0 [points]
36	48 [ways]
37	22 [times]
38	14 [= 100th term]
39	16
40	5 [white pairs]

5th grade — May 16,
2026

"Math Is Cool" Masters — 2025-26

Total Correct (all columns)

Room #

SCHOOL NAME

STUDENT NAME

Team #

Individual Contest - Score Sheet

STUDENTS: DO NOT WRITE IN SHADED REGIONS

	Answer	1 or 0	1 or 0
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
1-15 TOTAL:			

	Answer	1 or 0	1 or 0
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
16-30 TOTAL:			

	Answer	1 or 0	1 or 0
31			
32			
33			
34			
35			
36			
37			
38			
39			
40			
31-40 TOTAL:			

5th grade — May 16, 2026

Scorers: Just score as 0 or 1 and add up those values (i.e., just work with number correct).

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Team Multiple Choice Contest

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #4.

Each cell of this 6x6 grid contains an integer. Some of these integers are not revealed to you, and a couple of them are indicated by symbols. The numbers in each column and each row form an arithmetic sequence. An arithmetic sequence is a list of numbers where the same number is added or subtracted each time to get the next number. For example, 2, 5, 8, 11, 14 is an arithmetic sequence because the number 3 is being added each time.

1				45	#
					48
	4				
-11					
?			4		

- 1 What is the value of the integer marked with a "?"
A) -14 B) -16 C) -18 D) -20 E) Answer not given.
- 2 What is the mean (average) of all the integers in the third column from the left?
A) 6 B) 10.5 C) 12 D) 12.5 E) Answer not given.
- 3 What is the sum of all the integers marked in the four shaded cells?
A) 64 B) 71 C) 75 D) 79 E) Answer not given.
- 4 In the grid above, 6 of the integers were revealed. What is the fewest number of integers that could be revealed where a student could still determine the identity of every other integer in the grid?
A) 3 B) 4 C) 5 D) 6 E) Answer not given.

Continued on Next Page

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #5 THROUGH #7.

The sport of figure skating has jumps that are scored with the following table:

Note: Abbreviations for the jumps are written with the number of revolutions first, then the jump type. For example, 3T would be a triple toe loop with 3 revolutions and 4A would be a quadruple axel with 4 revolutions.

POINTS FOR EACH JUMP, BY NUMBER OF REVOLUTIONS

JUMP TYPE	Single (1)	Double (2)	Triple (3)	Quadruple (4)
TOE LOOP (T)	0.40	1.30	4.20	9.50
SALCHOW (S)	0.40	1.30	4.30	9.70
LOOP (L)	0.50	1.70	4.90	10.50
FLIP (F)	0.50	1.80	5.30	11.00
LUTZ (LZ)	0.60	2.10	5.90	11.50
AXEL (A)	1.10	3.30	8.00	12.50

5 If Ilia does two triple flips, three triple toe loops, one quadruple lutz and one quadruple axel in his long program, how many total revolutions has he made?

- A) 18 B) 21 C) 23 D) 24 E) Answer not given.

6 A common Olympic-level jump combination is a triple flip (3F) followed by a triple toe loop (3T) followed by a double toe loop (2T). What is the total value of this combination, using the values in the table above?

- A) 10.40 B) 10.50 C) 10.60 D) 10.90 E) Answer not given.

7 A single Euler is another type of jump that can create combinations by coming in between two different jumps. How many possible three-jump combinations with three different jumps have the single Euler as the second jump? Jumps are different if they either have a different type or a different number of revolutions. For example, a double axel (2A) is different from a triple axel (3A).

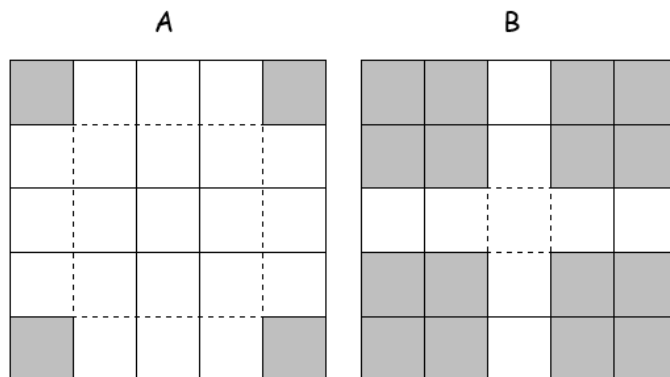
- A) 529 B) 552 C) 576 D) 1656 E) Answer not given.

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USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #8 THROUGH #10.

A square grid composed of unit squares (side length = 1 unit) can be used to construct different sized boxes, with a square base and no top, by removing squares from each corner of the grid and folding up the sides.

For example, in a 5x5 square grid, either 1x1 squares can be cut out of each corner (A), or 2x2 squares can be cut out of each corner (B). In each case, the remaining sides can be folded upwards along the dotted lines to create a box.



- 8** In example B shown above, what is the volume in cubic units of the box created when the four sides are folded up?
- A) 1 cubic units B) 2 cubic units C) 4 cubic units D) 5 cubic units E) None of the above.
- 9** Instead of starting with a 5x5 grid as shown in the example, start with an 18x18 grid composed of unit squares. By cutting squares out of the corners of the 18x18 grid and folding up the sides, how many different sized boxes can be constructed?
- A) 6 B) 7 C) 8 D) 9 E) None of the above.
- 10** What is the largest volume of a box that can be created in this manner by starting from an 18x18 grid?
- A) 392 cubic units B) 400 cubic units C) 432 cubic units D) 452 cubic units
E) None of the above.

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5th grade — May 16, 2026

Sponsored by: WEA - Eastern

Key

Team Multiple Choice Contest - Answer Key

5th grade

Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

Answer	
1	A
2	B
3	D
4	B
5	C
6	E [10.80]
7	B
8	B
9	C
10	C

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5th grade — May 16, 2026

Final Score (out of 20)

Room #

School Name

Team #

Team Multiple Choice Contest - 15 minutes - ~20% of team score

You will have 15 minutes to answer 10 multiple choice questions as a team. This test is the only test where you will be penalized for incorrect responses. You will receive two points for a correct letter response, zero points for leaving it blank, and minus one point for an incorrect response. When you are prompted to begin, tear off the colored answer sheet, pass out a copy of the test to each team member, and begin testing. **ONLY a letter response should be listed as an answer on this answer sheet.** Correct responses are worth 2 points, incorrect responses are worth -1 point, and absence of a response is worth 0 points.

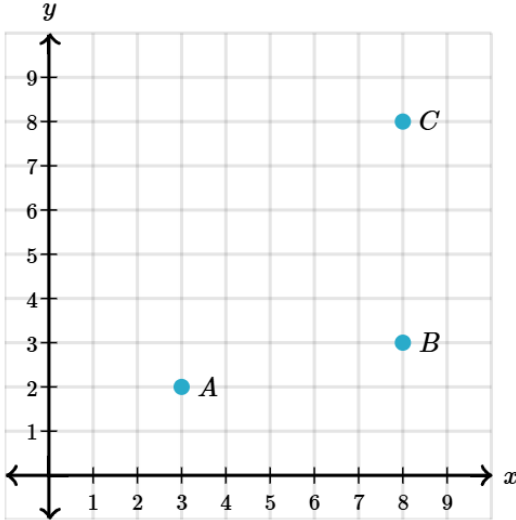
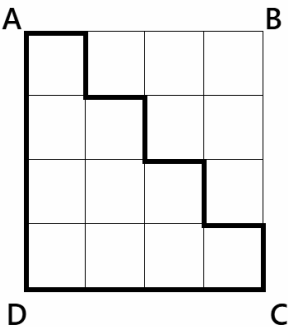
STUDENTS: DO NOT WRITE IN SHADED REGIONS

		Scorer 2	Scorer 1
Answer		-1, 0, or 2	-1, 0, or 2
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5th grade	TOTAL:		

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Team Contest

1	Finley is supposed to get off work at 5 P.M., but she works overtime until 7:42 P.M. that day. How many extra minutes did Finley work?
2	How many units away from the y-axis is point C? 
3	When the following multiplication is done, how many zeros are at the end of the number? $100 \times 5 \times 3 \times 2 \times 1$
4	Square ABCD is made up of smaller congruent (equal) squares. Square ABCD has a perimeter of 32 units. In units, what is the perimeter of the portion of the square with the darkened outline? 
5	Biff has 2 one-dollar bills, 3 quarters, 2 dimes and 6 pennies. Then Eho gives him 5 nickels. How many total cents does Biff have now?
6	In this year's Science Bowl competition, Pullman High School played Moscow High School 3 times. The final scores in points were 138 - 20, 104 - 48, and 76 - 22, with Pullman winning all 3. In points, what was Pullman's average margin of victory against Moscow over the three rounds?
Continued on next page.	

7	Camden's pin number is a 4-digit positive integer that is less than 5000. It contains the digits 1, 3, 6 and 8 in some order. The 1 is not next to the 3, the 6 is not next to the 1, and the 3 is next to the 8. What is Camden's pin number?
8	The sum of the digits in the numbers 52 and 53 is $(5 + 2) + (5 + 3) = 7 + 8 = 15$. Find the sum of the digits in the first 35 counting numbers, 1 through 35.
9	Izabel wants to draw an 10×10 grid made of unit squares. She has a square stamp that can draw one square. What is the smallest number of times she must use the stamp to create the entire grid?
10	Morse code uses a series of dots and dashes to represent each letter of the alphabet. There are two numbers spelled out in Morse code below that have been written without putting any spaces between the letters. What is the value of the larger number minus the smaller number?

A ● —	J ● — — —	S ● ● ●
B — ● ● ●	K — ● —	T —
C — ● — ●	L ● — ● ●	U ● ● —
D — ● ●	M — —	V ● ● ● —
E ●	N — ●	W ● — —
F ● ● — ●	O — — —	X — ● ● —
G — — ●	P ● — — ●	Y — ● — —
H ● ● ● ●	Q — — ● —	Z — — ● ●
I ● ●	R ● — ●	

Number 1:

● ● — ● — — — ● — ● — — ● — — ● ● ● ● — ● ● —

Number 2:

— ● ● ● ● ● ● — ● — ● ● — ●

"Math Is Cool" Masters — 2025-26

5th grade — May 16, 2026
Sponsored by: WEA - Eastern

Key

Team Contest - Answer Key

5th grade

Answer	
1	162 [minutes]
2	8 [units]
3	3 [zeroes]
4	32 [units]
5	326 [cents]
6	76 [points]
7	1836 [= pin number]
8	198 [= sum]
9	68 [times]
10	33

"Math Is Cool" Masters — 2025-26
5th grade — May 16, 2026

Final Score (out of 10)

Room #

School Name

Team #

Team Contest - 15 minutes - ~30% of team score

When you are prompted to begin, tear off the colored answer sheet and give a copy of the test to each of your team members and begin testing. Each problem is scored as a 1 or 0. Record all answers on this colored answer sheet.

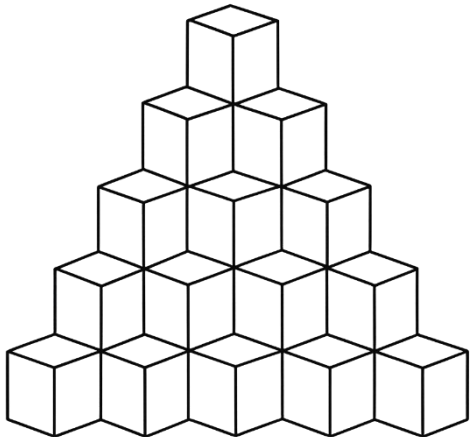
STUDENTS: DO NOT WRITE IN SHADED REGIONS

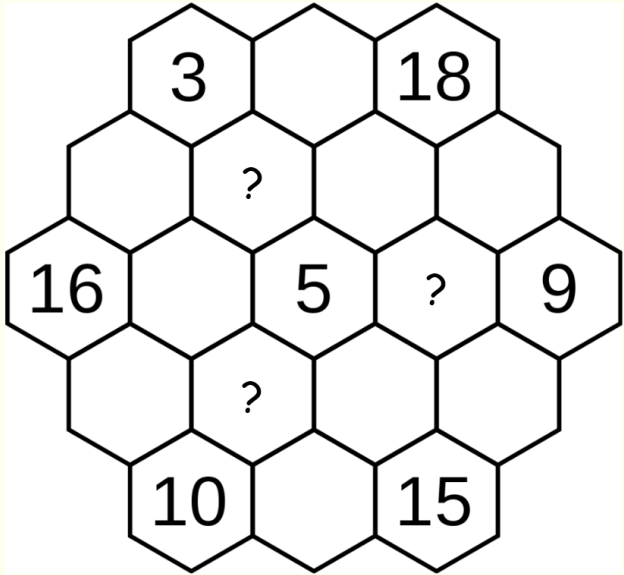
Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5th grade		TOTAL:	

"Math Is Cool" Masters — 2025-26

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Linda Moore Triple Jump

1	Four hours and 36 minutes equals how many minutes?
2	Evaluate: $24 \div (4 \times 2 - 5)$
3	In cubic centimeters, what is the volume of a rectangular prism that has a base area of 48 square centimeters and a height of $4\frac{1}{2}$ centimeters?
4	The largest stadium in the world is in North Korea. It covers 50 acres and is 197 feet tall, holding up to 114,000 spectators. What is the average number of spectators held per acre of this stadium?
5	Identical unit cubes are stacked against the walls in a corner of a room. How many of the cubes in the stack are not visible? 
6	Brandon runs along a straight road at a constant rate of 7 miles per hour. Two hours later, Liandro starts riding his bike in the same direction at a constant rate of 21 miles per hour. What is the total distance that Brandon runs, in miles, before Liandro catches up with him?
7	What even integer between 700 and 800 is divisible by both 5 and 13?
Continued on next page.	

8	In this magic hexagon, all of the integers 1 through 19 should be placed one per hexagon so that the sum along every straight line in any direction is always equal to 38. The rows will consist of 3, 4 or 5 hexagons. Some numbers have already been placed. What is the sum of the three numbers in the hexagons that are indicated with a question mark? 
9	The set of integers: 5, 7, 12, X, Y, has the same median, unique mode, and range. What is the sum of X and Y, if $X > Y > 6$?
10	A sequence of numbers begins as follows. Starting with the third term, each term is equal to the term preceding it minus the term preceding that one. What is the sum of the first 2026 terms of the sequence? 1, 3, 2, -1, ...

"Math Is Cool" Masters — 2025-26

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Sponsored by: WEA - Eastern

Key

Linda Moore Triple Jump - Answer Key

5th grade

Answer	
1	276 [minutes]
2	8
3	216 [cm ³]
4	2280 [spectators]
5	20 [cubes]
6	21 [miles]
7	780
8	17
9	29 [= X + Y]
10	5 [= sum]

"Math Is Cool" Masters — 2025-26

5th grade — May 16, 2026

Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

When you are prompted to begin, tear off the three colored answer sheets and give a copy of the test to each of your team members and begin testing. Record all answers on this colored answer sheet. This Submittal #1 will be collected after 5 minutes.

SUBMITTAL #1

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th grade	TOTAL:		

"Math Is Cool" Masters — 2025-26

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Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #2 will be collected after 10 minutes.

SUBMITTAL #2

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th grade	TOTAL:		

"Math Is Cool" Masters — 2025-26
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Final Score (out of 10)

Room #

School Name

Team #

Linda Moore Triple Jump - 15 minutes - ~15% of team score

This Submittal #3 will be collected after 15 minutes.

SUBMITTAL #3

STUDENTS: DO NOT WRITE IN SHADED REGIONS

Answer		Scorer 2	Scorer 1
		0 or 1	0 or 1
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
5 th grade	TOTAL:		

"Math Is Cool" Masters — 2025-26

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Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Masters — 2025-26

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Room #

School Name

Team #

Total Score for Each Round

College Bowl #1 (10 Possible)	College Bowl #2 (10 Possible)	College Bowl #3 (10 Possible)

DO NOT USE TALLY MARKS ON THIS SHEET. WRITE THE TOTAL SCORE FOR EACH ROUND.

"Math Is Cool" Masters — 2025-26

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Proctor
Copy

Mental Math Contest

MENTAL MATH - 30 seconds per question - ~25% of team score & ~8% of individual score

All students in the room will concurrently be asked the same eight questions in this individual test. When it is time to begin, the proctor will read the first question twice. You may not do any writing or talking while arriving at a solution. Once you have a solution, record it on the sheet in front of you. **You may not change or cross out answers once you have written an answer down. If there are eraser marks, write-overs, or crossed-out answers, they will be marked wrong.** Once all students have laid their pencils on the desk, another question will be asked. If a student doesn't lay his or her pencil down, the maximum wait time is 30 seconds after completion of the second reading of the question before the next question is read. You may continue to work on a problem (in your head) while the next question is being read. The raw score is 1 point per correct answer.

1	In meters, what is the perimeter of a square with an area of nine square meters?	
2	Playing ping pong for one hour burns one hundred fifty calories. How many calories would you burn playing ping pong for one hundred twenty minutes?	
3	What is the range of the following set: twelve, seventeen, fifteen, twenty-eight	
4	What is the third smallest positive integer factor of one hundred five?	
5	A hot air balloon took off with a sheep, a duck, and a rooster on board. However, they had to board one at a time. In how many different orders could the animals have boarded the balloon?	
6	Solve this equation for x: Three X plus two X plus five plus seven equals twenty-seven	
7	The ratio of dogs to cats at an animal shelter was seven to five. After twelve dogs and zero cats were adopted out, the ratio of dogs to cats is now one to one. How many total dogs and cats were at the shelter originally?	
8	A lion's roar can be heard from five miles away. From how many yards away can a lion's roar be heard?	

"Math Is Cool" Masters — 2025-26

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Key

COLLEGE BOWL ROUND #1

#	Problem	Answer
1	What is nine squared?	81
2	A car travels two hundred seventy miles in four and a half hours. What is its average speed in miles per hour?	60 [mph]
3	A rectangle has a perimeter of fifty-four centimeters and a length of sixteen centimeters. What is the area of the rectangle in square centimeters?	176 [sq cm]
4	Three consecutive odd integers have a sum of eighty-one. What is the largest of the three integers?	29
5	A jar contains red and blue marbles in a ratio of three to five. There are forty marbles in total. How many more blue marbles are there than red marbles?	10 [marbles]
6	What is the next palindrome year after the year two thousand two? A palindrome reads the same forwards and backwards.	2112
7	A pump fills a tank at fifteen gallons per minute, while a drain empties it at nine gallons per minute. The tank holds ninety gallons. Starting empty, how many minutes will it take to fill the tank?	15 [minutes]
8	What is the sum of all positive factors of twenty-eight?	56
9	How many of the following numbers are divisible by eleven: Forty-four, sixty-six, eighty-eight, ninety-nine and one hundred ten.	5 [numbers]
10	What is four thousand fifteen divided by eleven?	365

"Math Is Cool" Masters — 2025-26

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Key

COLLEGE BOWL ROUND #2

#	Problem	Answer
1	What is the value of seven squared plus seven?	56
2	A recipe calls for three-quarters of a cup of sugar per batch. How many cups of sugar are needed for four batches?	3 [cups]
3	A right triangle has legs of length six and eight units. What is its area in square units?	24 [sq units]
4	The sum of two numbers is fifty and their difference is fourteen. What is the larger of the two numbers?	32
5	In how many ways can three students be chosen from a group of six to form a committee?	20 [ways]
6	How many positive integers less than one hundred are divisible by five?	19 [integers]
7	Train A leaves a station at nine AM traveling at sixty miles per hour. Train B leaves the same station at ten AM traveling in the same direction at ninety miles per hour. How many hours does Train B travel before it catches Train A?	2 [hours]
8	What is the measure of each interior angle of a regular hexagon in degrees?	120 [degrees]
9	What is the greatest common factor of seventy-two and ninety?	18 [= gcf]
10	Twelve times four equals what number times three?	16

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Key

COLLEGE BOWL ROUND #3

#	Problem	Answer
1	What is the value of one hundred minus the square root of one hundred?	90
2	How many square centimeters are in two square meters?	20,000 [sq cm]
3	Lily counts up by fives. The seventh number she says is forty. What was the third number she said?	20
4	If A equals ninety-nine, B equals one hundred ninety-nine and C equals two hundred ninety-nine, what is the value of A plus B plus C?	597 [= A + B + C]
5	The mean of six numbers is fifteen. When two of the numbers are removed, the mean of the remaining four is twelve. What is the mean of the two removed numbers?	21 [= mean]
6	The digits one, two, three and four can be used exactly once each to write a certain number of four-digit numbers. How many of these four-digit numbers are even?	12 [numbers]
7	In centimeters, what is the radius of a circle that has a circumference of twenty-two pi centimeters?	11 [cm]
8	What is the sum of the first ten positive even numbers?	110 [= sum]
9	What is the least common multiple of eight and eighteen?	72 [= LCM]
10	What is twenty percent of two hundred fifty?	50

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Key

COLLEGE BOWL ROUND #4

#	Problem	Answer
1	What is the value of three to the fourth power minus four to the third power?	17
2	How many minutes are in three days?	4,320 [minutes]
3	A square and a regular pentagon have the same perimeter. The side length of the pentagon is eight inches. In square inches, what is the area of the square?	100 [sq cm]
4	There are two cats in front of a cat, two cats behind a cat and a cat in the middle. What is the fewest number of cats there could be?	3 [cats]
5	How many different three-digit numbers can be formed using the digits two, five and eight if each digit is used exactly once?	6 [numbers]
6	A sequence starts with two, and each term equals double the previous term plus three. What is the fourth term?	37 [= 4 th term]
7	An integer is chosen at random from the integers one through ten inclusive. What is the probability in percent that it contains the digit one?	20 [%]
8	A right triangle has legs of nine and twelve centimeters. What is its perimeter in centimeters?	36 [cm]
9	How many of the following are perfect squares? Sixteen, thirty-six, fifty, eighty-one, one hundred, and one hundred forty-four.	5 [numbers]
10	Lennon can read three thousand seven hundred fifty words in fifteen minutes. On average, how many words does he read per minute?	250 [words]

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Key

COLLEGE BOWL ROUND #5

#	Problem	Answer
1	How many quarters are equal in value to fifteen dollars and seventy-five cents?	63 [quarters]
2	How many grams are in two point three five kilograms?	2350 [g]
3	If it takes five workers twelve days to complete a job, how many days would it take ten workers to complete the same job?	6 [days]
4	A twenty-five percent discount is applied to a one hundred twenty dollar item, then an additional ten percent loyalty discount is applied. What is the final price in dollars?	81 [\$]
5	How many multiples of one hundred are between one and two thousand twenty-six?	20 [multiples]
6	Ella has twelve different colored shirts and ten different colored jackets. In how many ways can she pick one shirt and one jacket assuming that she does not want to wear her blue shirt?	110 [ways]
7	A sequence starts with three, seven, fifteen, thirty-one, sixty-three, and so on. What is the next term?	127 [= next term]
8	What is the side length in units of a regular nonagon with a perimeter of one hundred forty-four units?	16 [units]
9	What is the median of the following set of numbers: one, thirty-three, nine hundred ninety-nine, twelve, sixteen, zero	14 [= median]
10	In the sequence one hundred, ninety-one, eighty-two, seventy-three and so on, what is the first term less than fifty?	46

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Key

COLLEGE BOWL ROUND #6

#	Problem	Answer
1	Brian has twenty marbles, Ernie has three times as many marbles as Brian, and Navani has eight more marbles than Ernie. How many marbles do they have altogether?	148 [marbles]
2	How many quarts are in three and a half gallons?	14 [quarts]
3	How many edges does a cube have?	12 [edges]
4	A circle is split into eight equal slices. How many slices have to be colored in to represent three-fourths of the circle?	6 [slices]
5	Seven students scored the following on a quiz: eight, eleven, fourteen, seventeen, twenty, twenty-five, and one unknown score N. If the median is fourteen, what is the largest possible integer value of N?	14 [points] [N]
6	What is the largest prime factor of eighty-four?	7
7	Two boats leave the same dock at the same time. One travels north at eight miles per hour, the other travels east at six miles per hour. How many miles apart are they after three hours?	30 [miles]
8	What is the value of the square root of one hundred forty-four plus the square root of two hundred twenty-five?	27
9	Four over twenty-four equals twelve over what number?	72
10	A rectangular tile measures two inches by three inches. What is the smallest number of tiles needed to completely cover a two foot by two foot square region?	96 [tiles]

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Key

COLLEGE BOWL - EXTRA QUESTIONS

#	Problem	Answer
1	A cyclist rides four miles in twenty minutes. At the same rate, how many miles does she ride in one and a half hours?	18 [miles]
2	If a square has an area of forty-nine square units, what is the value in units of its perimeter times two?	56 [units]
3	There are two green marbles and eight red marbles in a bag. If one marble is randomly selected, what is the probability in percent that it is not green?	80 [%]
4	Over six games, a player scored eighteen, twenty-two, fifteen, thirty, twelve, and twenty-five points. What was the median score in points?	20 [points] [median]
5	A right triangle has an area of thirty square centimeters and a base of twelve centimeters. What is the length of the hypotenuse in centimeters?	13 [cm]
6	What is fifteen percent of fifteen percent of four hundred?	9
7	What number, when you subtract seven and multiply the result by four, gives fifty-two?	20
8	Nine coins total seventy-five cents, using only dimes and nickels. How many dimes are there?	6 [dimes]
9	What is the mean of the following set: ninety, eighty-seven, ninety-three	90 [= mean]
10	How many seconds are in fifteen minutes?	900 [seconds]